

SCS Studio

Every selectable room type — furnished by the engine

2D plans with clearance zones + German ASR criteria · 3D X-ray renders

18 room types in this document:

• Office — Zellenbüro	5 × 4 m — 7 items placed
• Workspace — heavy storage	5 × 4 m — 7 items placed
• Meeting — Besprechungsraum	5 × 4 m — 11 items placed
• Presentation hall — Seminarraum	8 × 7 m — 46 items placed
• Reception — Empfang	5 × 4 m — 7 items placed
• Break room — Pausenraum	5 × 4 m — 7 items placed
• Quiet room — Ruheraum	3.5 × 3.5 m — 1 items placed
• Living room — Wohnzimmer	5 × 5 m — 9 items placed
• Lounge	4.5 × 4 m — 5 items placed
• Bedroom — Schlafzimmer	4 × 3.5 m — 2 items placed
• Dining — Esszimmer	4.5 × 4 m — 7 items placed
• Kitchen — Küche	5 × 4 m — 13 items placed
• Storage — Lager/Keller	4 × 3 m — 2 items placed
• Wardrobe — Garderobe	4 × 3 m — 6 items placed
• Entry — Foyer/Eingang	5 × 3 m — 7 items placed
• Balcony — Balkon	4 × 2.5 m — 5 items placed
• Print room — Kopierraum	3.5 × 3 m — 5 items placed
• IT/server — EDV-Raum	4 × 3.5 m — 4 items placed

Every layout: the engine's own suggestion (medium density), solved by the CP-SAT placer under ASR workplace rules and human placement logic. Hatched green = clearance the solver reserves; arrows = facing direction; dashed outlines = wall-/counter-mounted items.

Office — Zellenbüro

5 × 4 m · medium density · 7 items placed · 0 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

2 × **desk**

Workstation anchor — user side kept free, $\geq 1.5 \text{ m}^2$ movement area (ASR A1.2)

2 × **office chair**

Belongs to a desk or table, always FACING it

2 × **monitor**

ON the desk, screen facing the sitter

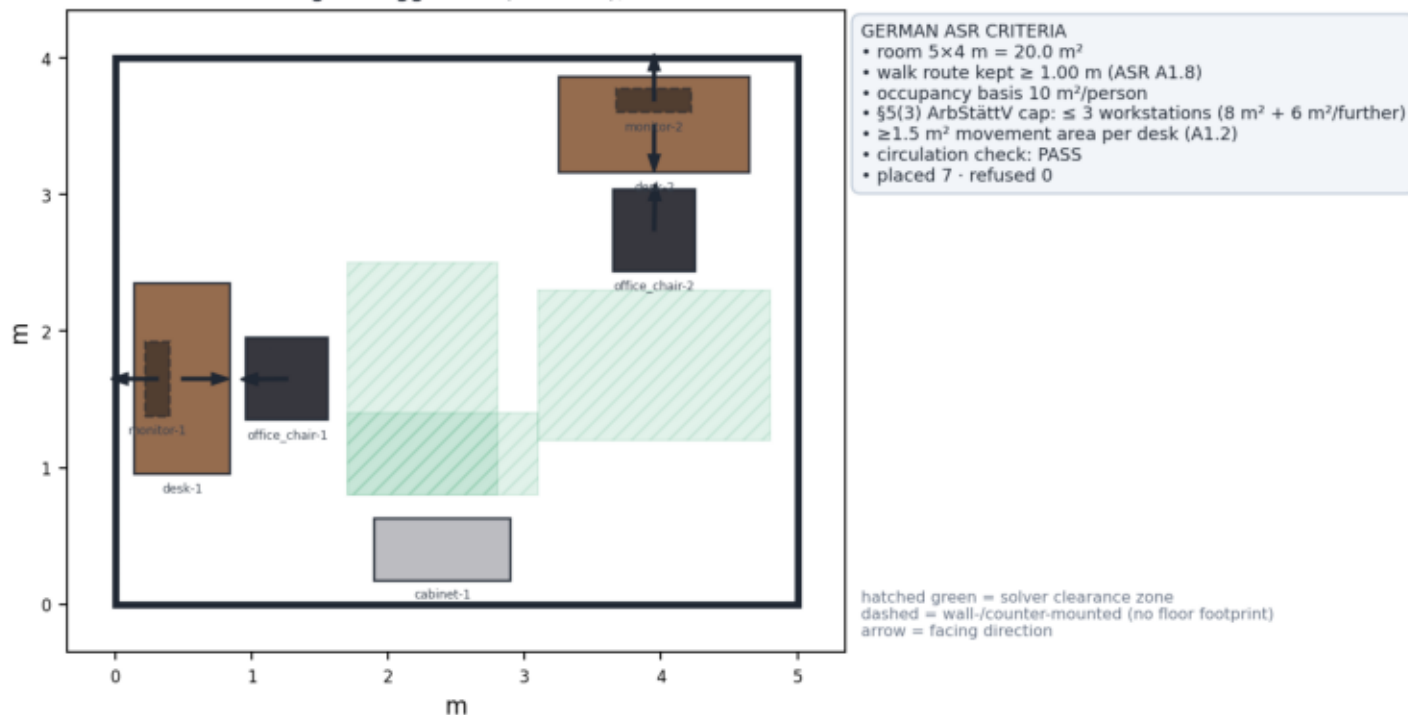
1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

Office — Zellenbüro

2D plan — clearance zones + ASR panel

Office — Zellenbüro — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Office — Zellenbüro — 7 pieces (X-ray)



Workspace — heavy storage

5 × 4 m · medium density · 7 items placed · 0 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

1 × **desk**

Workstation anchor — user side kept free, $\geq 1.5 \text{ m}^2$ movement area (ASR A1.2)

1 × **office chair**

Belongs to a desk or table, always FACING it

1 × **monitor**

ON the desk, screen facing the sitter

1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

1 × **waste bin**

At arm's reach BESIDE the desk/table ($\sim 0.2 \text{ m}$) — movable, minimal clearance

1 × **filing cabinet**

Perimeter storage beside the desks

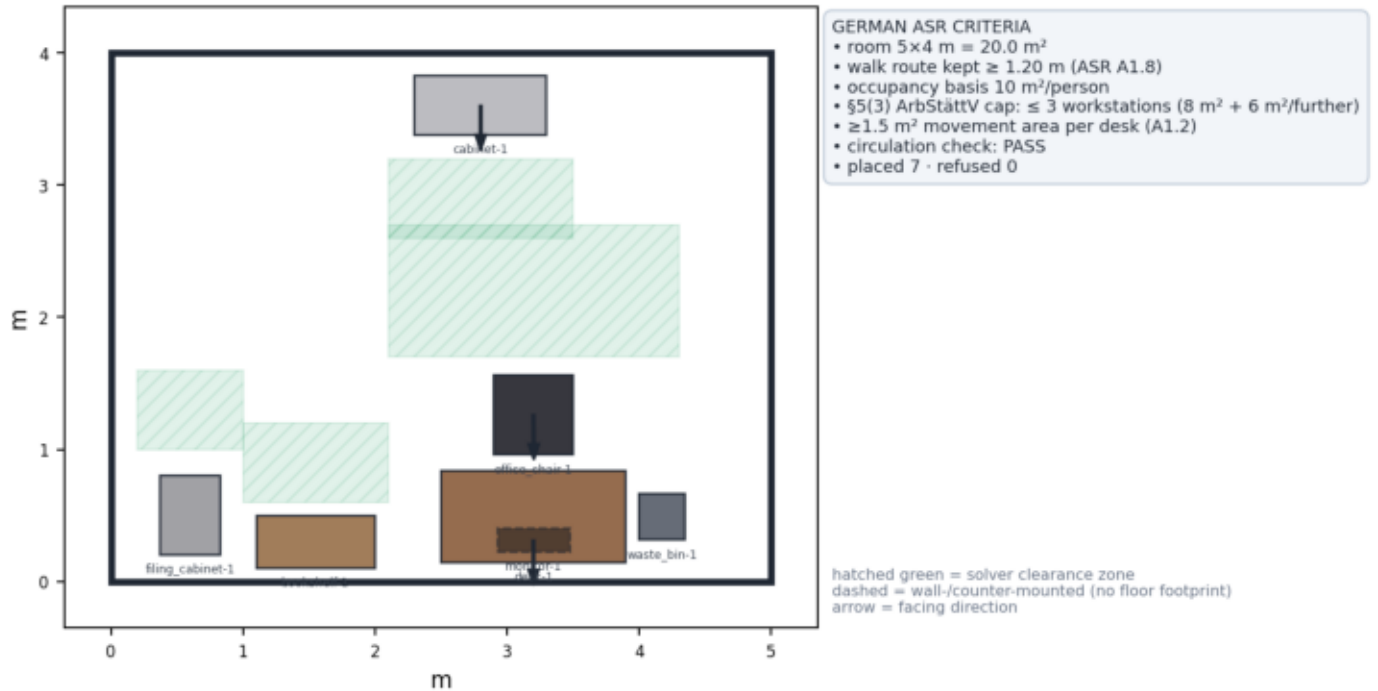
1 × **bookshelf**

Perimeter storage — flat against a wall

Workspace — heavy storage

2D plan — clearance zones + ASR panel

Workspace — heavy storage — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Workspace — heavy storage — 7 pieces (X-ray)



Meeting — Besprechungsraum

5 × 4 m · medium density · 11 items placed · 0 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

8 × **office chair**

Belongs to a desk or table, always FACING it

1 × **table**

Shared surface — chairs gather around it; rectangular tables seat the edges, never a 'petal' ring

1 × **flipchart**

Front corner, next to the display wall

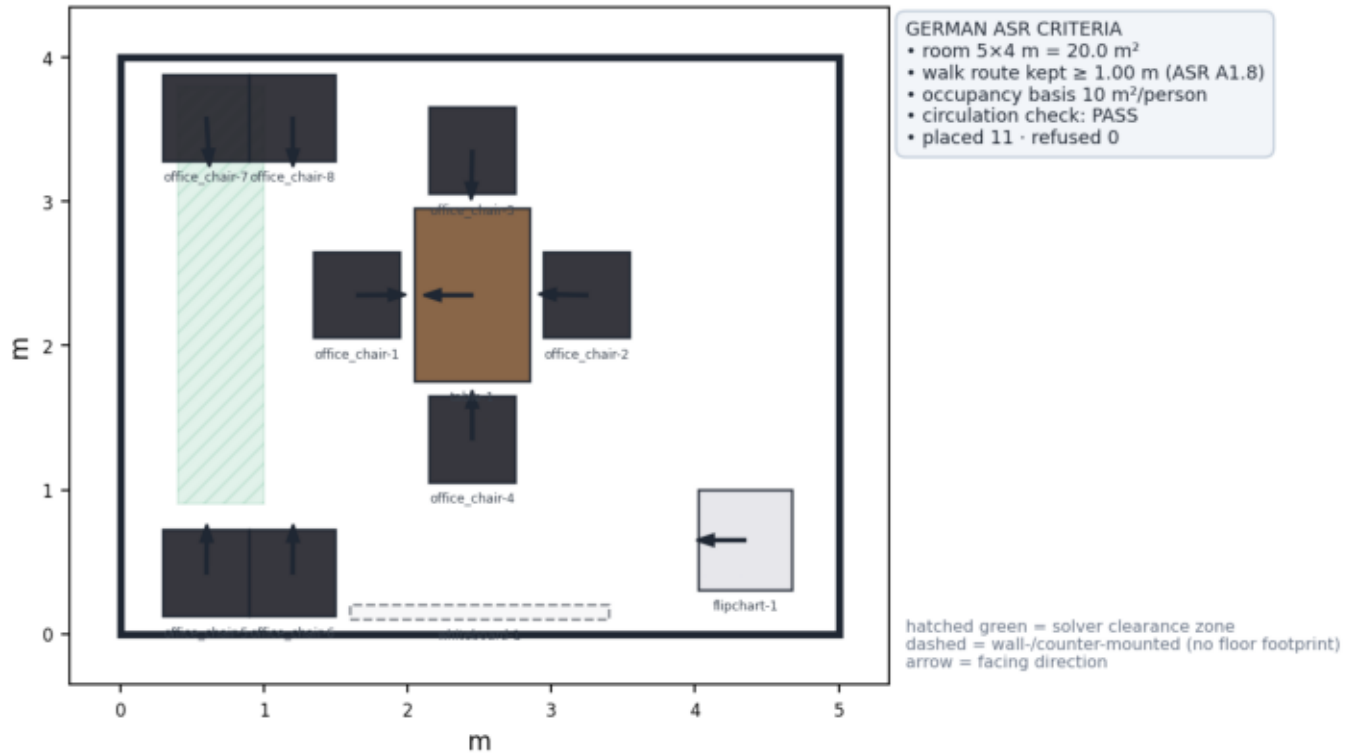
1 × **whiteboard**

Wall-mounted at 0.90 m, visibility strip kept clear

Meeting — Besprechungsraum

2D plan — clearance zones + ASR panel

Meeting — Besprechungsraum — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Meeting — Besprechungsraum — 11 pieces (X-ray)



Presentation hall — Seminarraum

8 × 7 m · medium density · 46 items placed · 0 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

40 × chair

Regular chair — at tables (max 4 per side, centered); spares park at the wall facing the room

1 × presentation screen

Owens the front wall (0.80 m mount), visibility strip kept clear

1 × lectern

Front of the room, beside the display axis

1 × projector

CEILING-mounted at 2.2 m, aimed at the screen, ASR-safe headroom

1 × whiteboard

Wall-mounted at 0.90 m, visibility strip kept clear

1 × flipchart

Front corner, next to the display wall

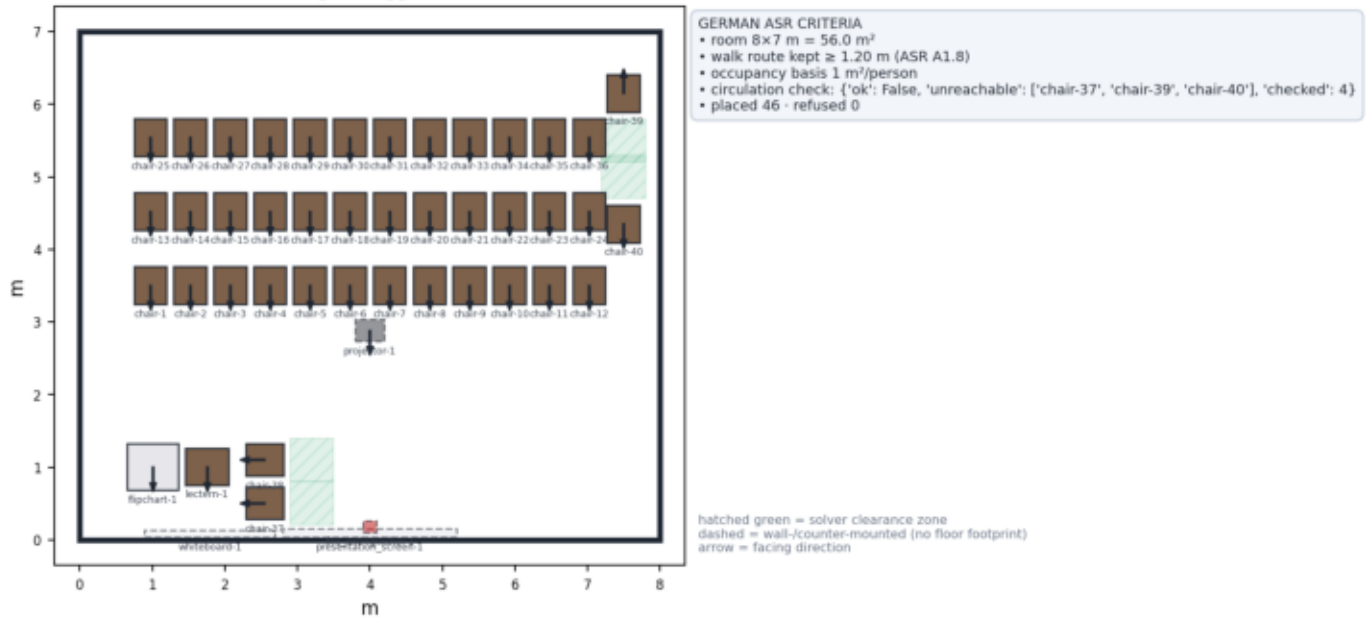
1 × fire extinguisher

Wall-mounted 1.00 m, PROTECTED access strip (ASR A2.2)

Presentation hall — Seminarraum

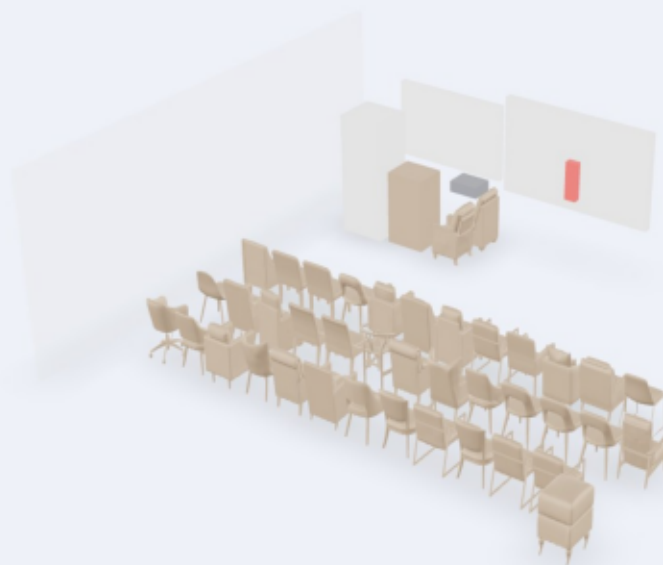
2D plan — clearance zones + ASR panel

Presentation hall — Seminarraum — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Presentation hall — Seminarraum — 46 pieces (X-ray)



Reception — Empfang

5 × 4 m · medium density · 7 items placed · 0 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

2 × **armchair**

In ROWS: 2 face the same way; 4 = two opposed pairs; never blocking the walkway

1 × **desk**

Workstation anchor — user side kept free, $\geq 1.5 \text{ m}^2$ movement area (ASR A1.2)

1 × **office chair**

Belongs to a desk or table, always FACING it

1 × **monitor**

ON the desk, screen facing the sitter

1 × **coat rack**

Near the door / entrance zone

1 × **side table**

At the end of a seating row / beside armchairs

1 × **planter**

Corners and window zones only — never mid-room

1 × **waste bin**

At arm's reach BESIDE the desk/table ($\sim 0.2 \text{ m}$) — movable, minimal clearance

1 × **water dispenser**

Perimeter, reachable from the aisle

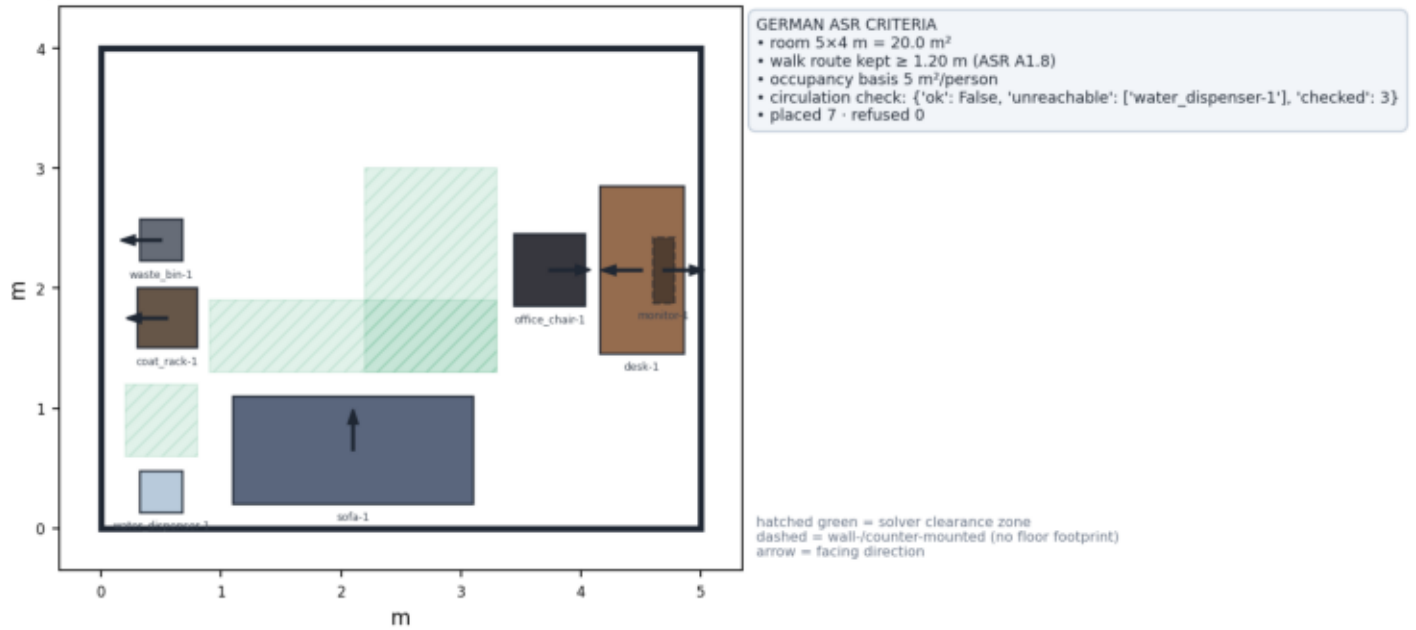
1 × **sofa**

Lounge anchor — coffee table in front, side table at the end

Reception — Empfang

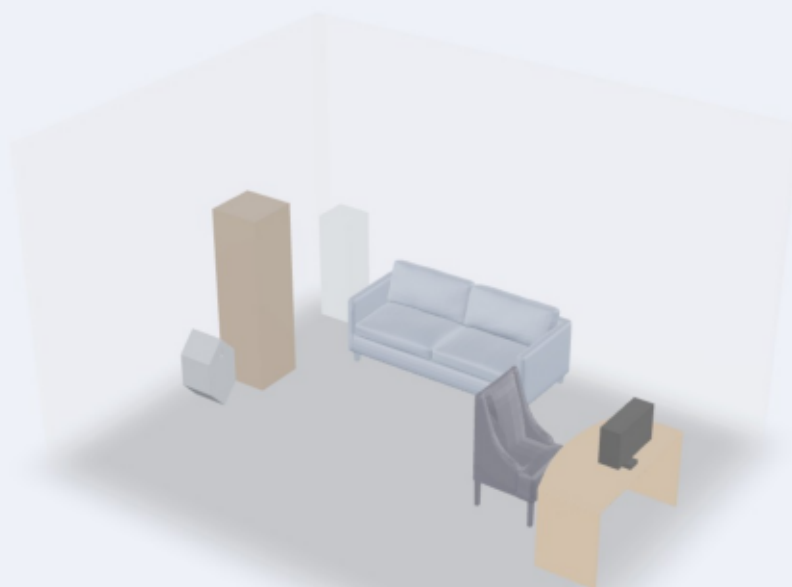
2D plan — clearance zones + ASR panel

Reception — Empfang — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Reception — Empfang — 7 pieces (X-ray)



Break room — Pausenraum

5 × 4 m · medium density · 7 items placed · 8 refused

WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

5 × **chair**

Regular chair — at tables (max 4 per side, centered); spares park at the wall facing the room

1 × **table**

Shared surface — chairs gather around it; rectangular tables seat the edges, never a 'petal' ring

1 × **coffee machine**

ON a counter or table

1 × **waste bin**

At arm's reach BESIDE the desk/table (~0.2 m) — movable, minimal clearance

1 × **water dispenser**

Perimeter, reachable from the aisle

1 × **planter**

Corners and window zones only — never mid-room

1 × **fridge**

Kitchen/break perimeter appliance

1 × **microwave**

ON a counter or table — never on the floor

1 × **first aid cabinet**

Wall-mounted 1.35 m, PROTECTED access strip (nothing may block it)

1 × **sofa**

Lounge anchor — coffee table in front, side table at the end

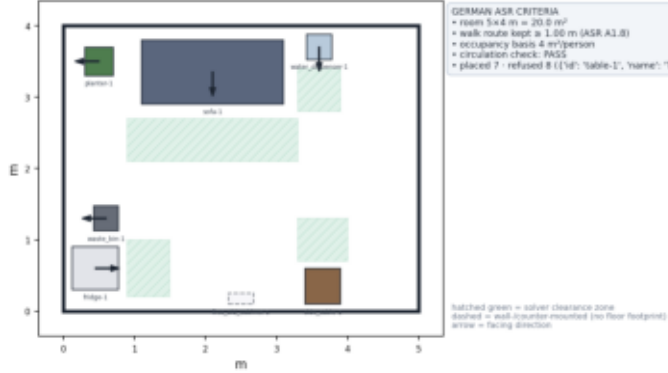
1 × **side table**

At the end of a seating row / beside armchairs

Break room – Pausenraum

2D plan — clearance zones + ASR panel

Break room — Pausenraum — engine suggestion (medium), clearance zones + ASR



GERMAN ASR CRITERIA
= mean 5 x 4 cm = 20.6 cm

- ```

GERMAN ASR CRITERIA
• resin 5x4 m = 20.5 m²
• walk route kept at 1.00 m (ASR A3.8)
• occupancy basis 4 m²/person
• circulation check PASS
• placed 7 - refuted 8 (['id': 'table-1', 'name': 'Table', 'category': 'table'], ['id': 'coffee_machine-1', 'name': 'Coffee Machine', 'category': 'coffee_machine'], ['id': 'chair-1', 'name': 'Chair', 'category': 'chair'])

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hatched green = silver clearance zone  
dashed = wall-jumper mounted (no floor footprint)  
arrow = facing direction

3D X-ray — shell ghosted, real placed pieces

Break room — Pausenraum — 7 pieces (X-ray)

# Quiet room — Ruheraum

3.5 × 3.5 m · medium density · 1 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### **2 × armchair**

In ROWS: 2 face the same way; 4 = two opposed pairs; never blocking the walkway

### **1 × side table**

At the end of a seating row / beside armchairs

### **1 × lamp**

Beside the reading seat

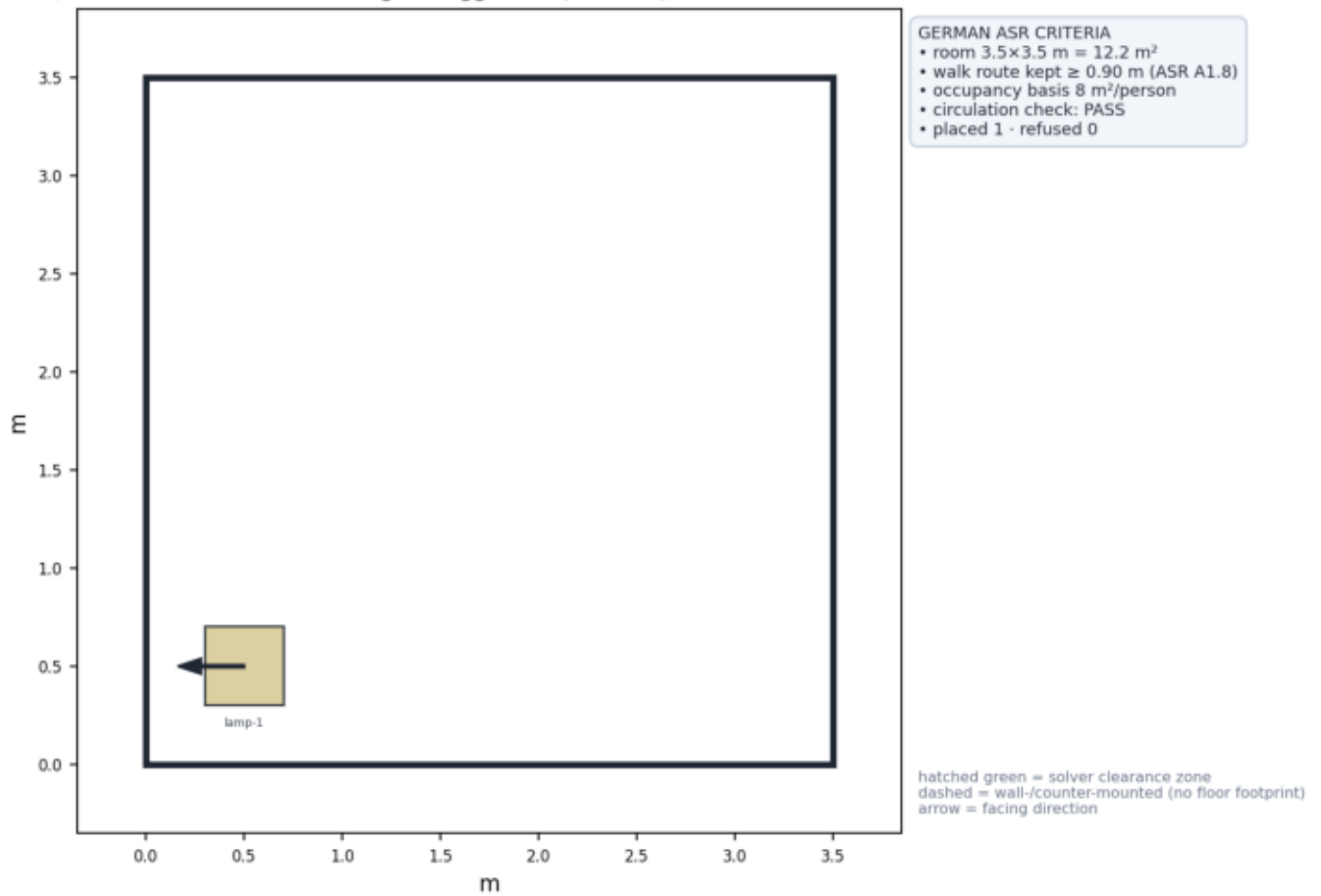
### **1 × planter**

Corners and window zones only — never mid-room

# Quiet room — Ruheraum

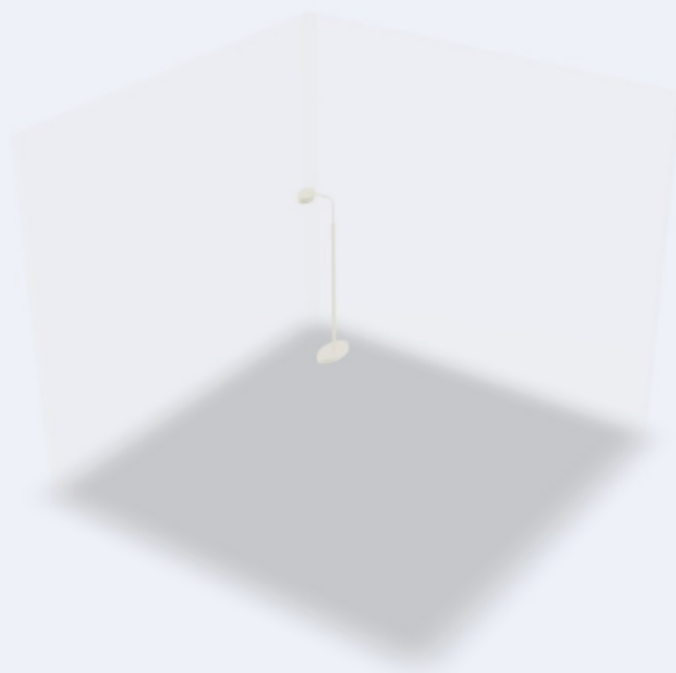
2D plan — clearance zones + ASR panel

Quiet room — Ruheraum — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Quiet room — Ruheraum — 1 pieces (X-ray)



# Living room — Wohnzimmer

5 × 5 m · medium density · 9 items placed · 4 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 3 × **chair**

Regular chair — at tables (max 4 per side, centered); spares park at the wall facing the room

### 2 × **stool**

Light seating at small tables / bar height

### 1 × **sofa**

Lounge anchor — coffee table in front, side table at the end

### 1 × **coffee table**

In front of the sofa, sitting distance

### 1 × **lamp**

Beside the reading seat

### 1 × **table**

Shared surface — chairs gather around it; rectangular tables seat the edges, never a 'petal' ring

### 1 × **armchair**

In ROWS: 2 face the same way; 4 = two opposed pairs; never blocking the walkway

### 1 × **side table**

At the end of a seating row / beside armchairs

### 1 × **bookshelf**

Perimeter storage — flat against a wall

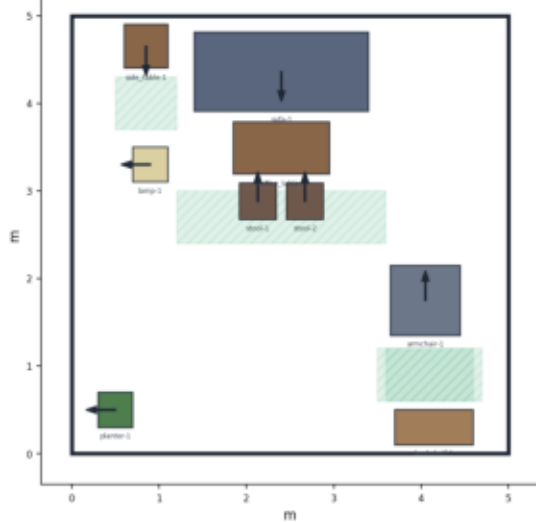
### 1 × **planter**

Corners and window zones only — never mid-room

# Living room — Wohnzimmer

## 2D plan — clearance zones + ASR panel

Living room — Wohnzimmer — engine suggestion (medium), clearance zones + ASR



GERMAN ASR CRITERIA

- room 5x5 m = 25.0 m<sup>2</sup>
- walk route kept ≥ 0.90 m (ASR A1.8)
- occupancy basis 4 m<sup>2</sup>/person
- circulation check: {'ok': False, 'unreachable': ['side\_table-1'], 'checked': 4}
- placed 9 - refused 4 (({'id': 'table-1', 'name': 'Table', 'category': 'table'}, {'id': 'chair-1', 'name': 'Chair', 'category': 'chair'}, {'id': 'chair-2', 'name': 'Chair', 'category': 'chair'})

hatched green = solver clearance zone  
dashed = wall counter-mounted (no floor footprint)  
arrow = facing direction

## 3D X-ray — shell ghosted, real placed pieces

Living room — Wohnzimmer — 9 pieces (X-ray)



# Lounge

4.5 × 4 m · medium density · 5 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **sofa**

Lounge anchor — coffee table in front, side table at the end

### 2 × **stool**

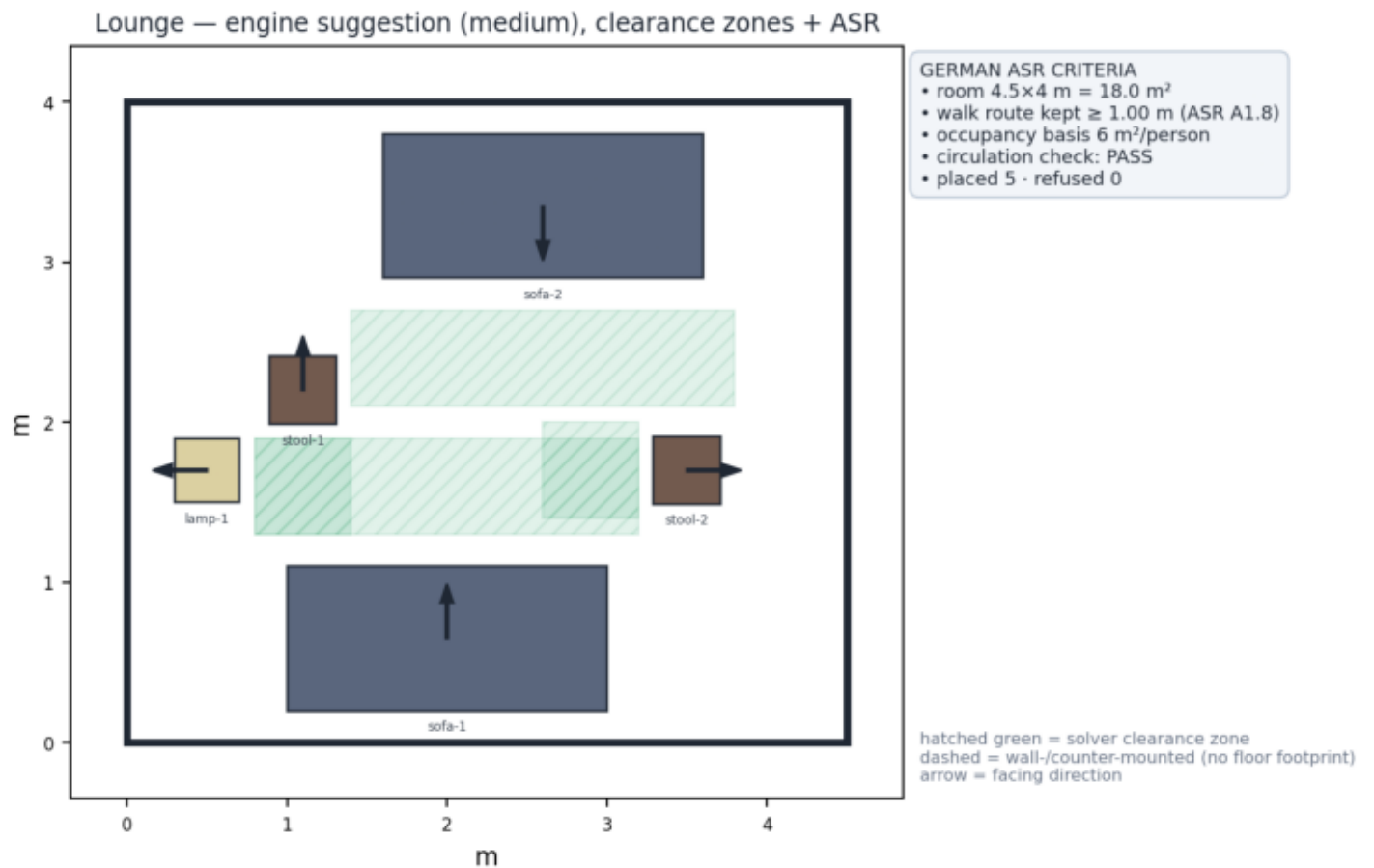
Light seating at small tables / bar height

### 1 × **lamp**

Beside the reading seat

# Lounge

## 2D plan — clearance zones + ASR panel



## 3D X-ray — shell ghosted, real placed pieces

Lounge — 5 pieces (X-ray)



# Bedroom — Schlafzimmer

4 × 3.5 m · medium density · 2 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

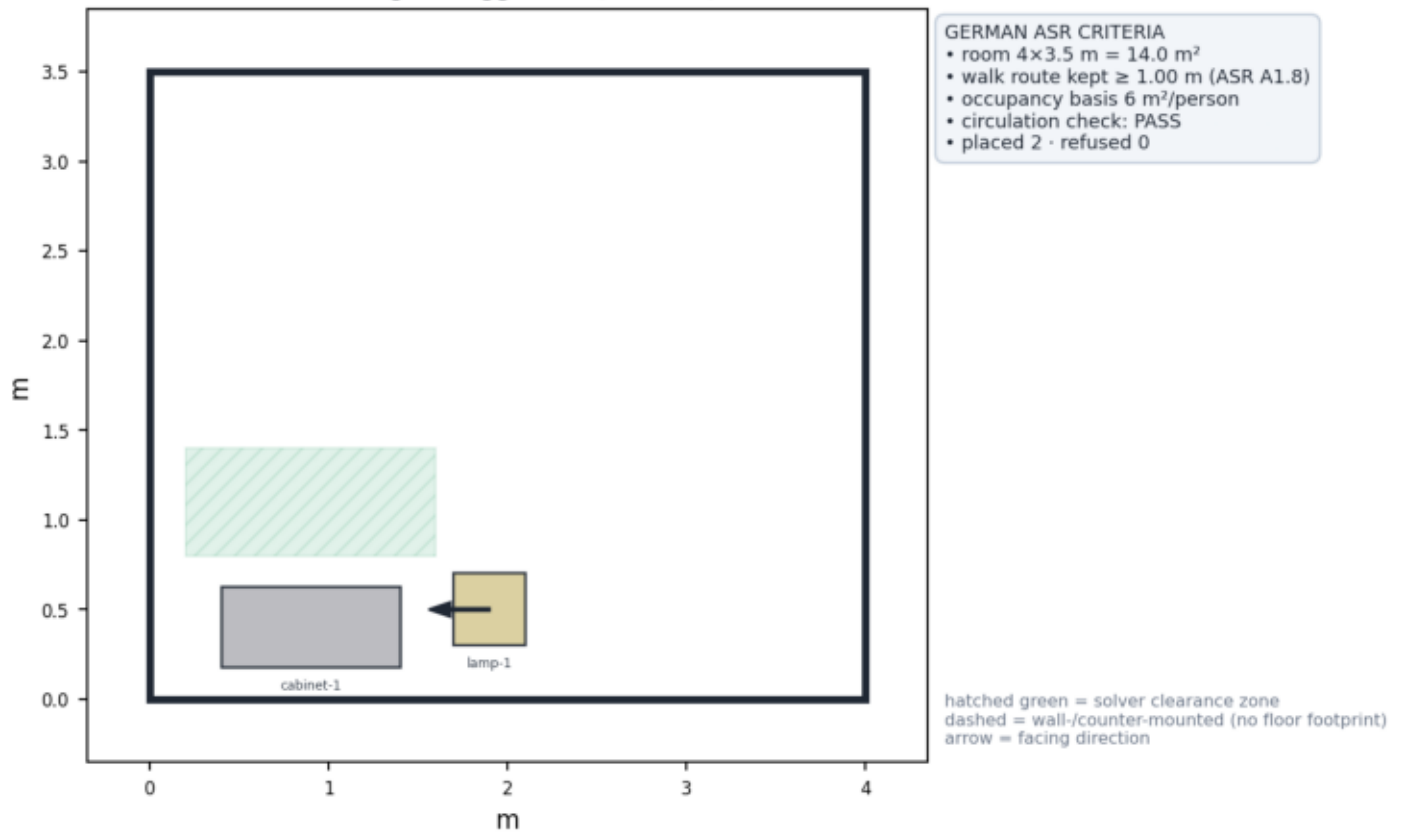
### 1 × **lamp**

Beside the reading seat

# Bedroom — Schlafzimmer

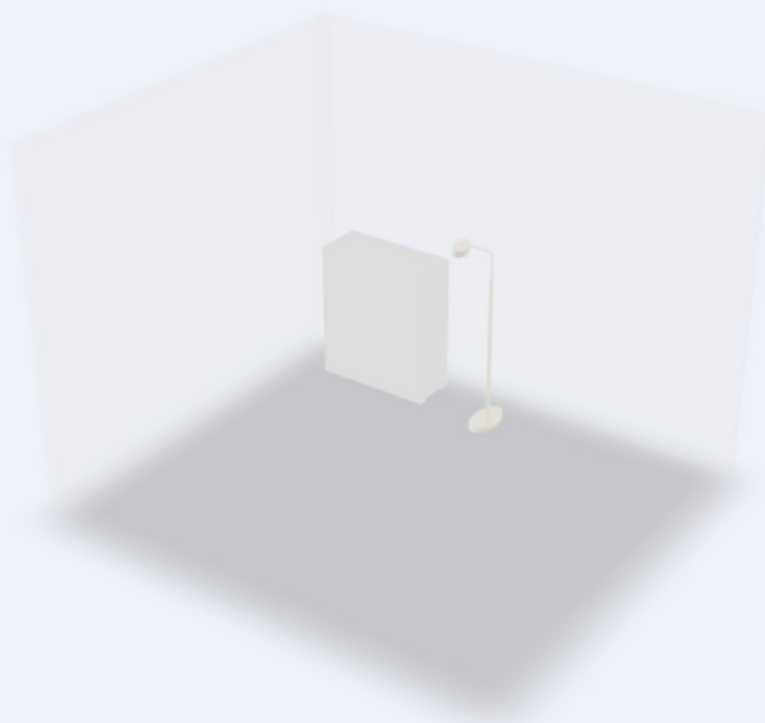
2D plan — clearance zones + ASR panel

Bedroom — Schlafzimmer — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Bedroom — Schlafzimmer — 2 pieces (X-ray)



# Dining — Esszimmer

4.5 × 4 m · medium density · 7 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 6 × **chair**

Regular chair — at tables (max 4 per side, centered); spares park at the wall facing the room

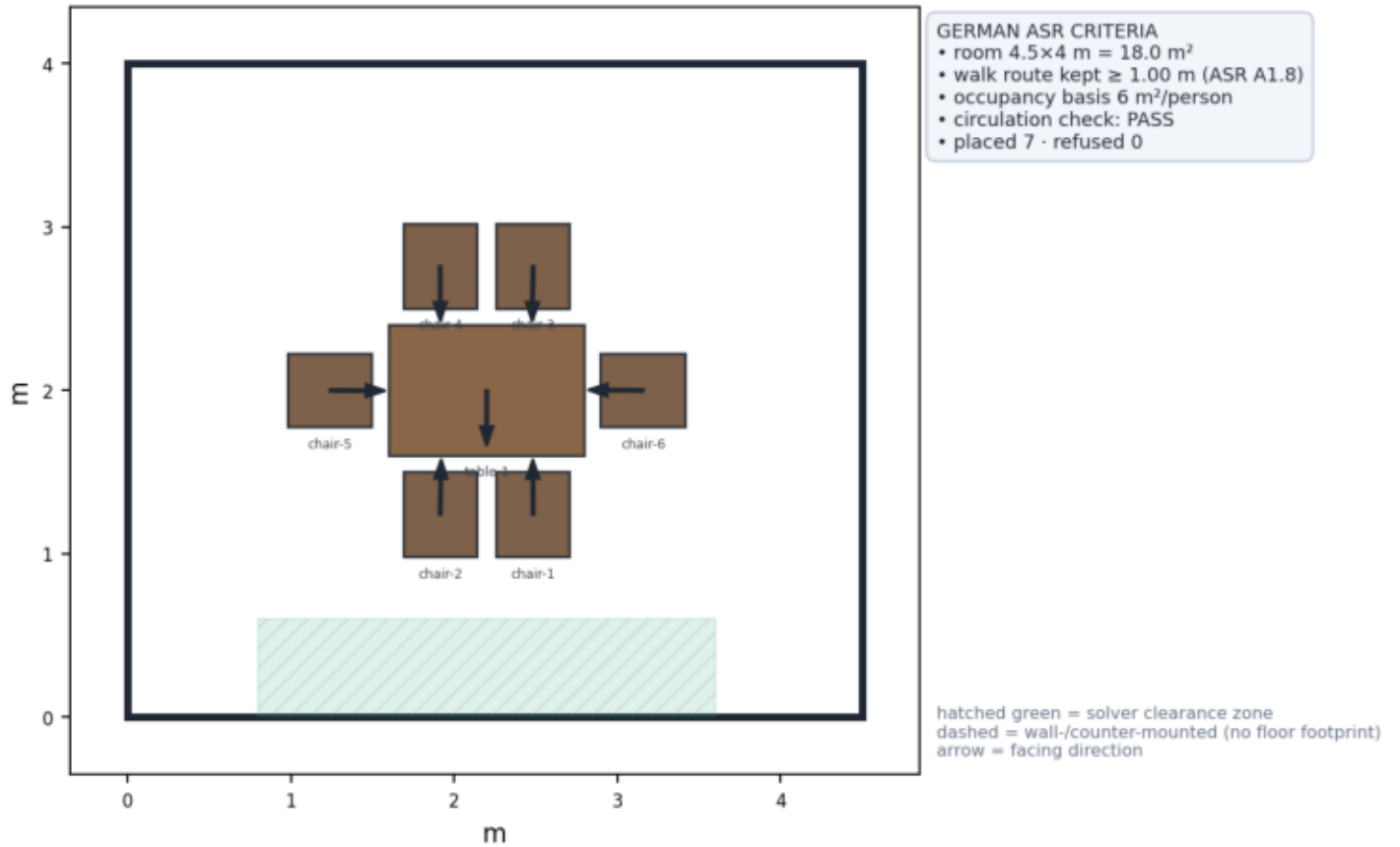
### 1 × **table**

Shared surface — chairs gather around it; rectangular tables seat the edges, never a 'petal' ring

# Dining — Esszimmer

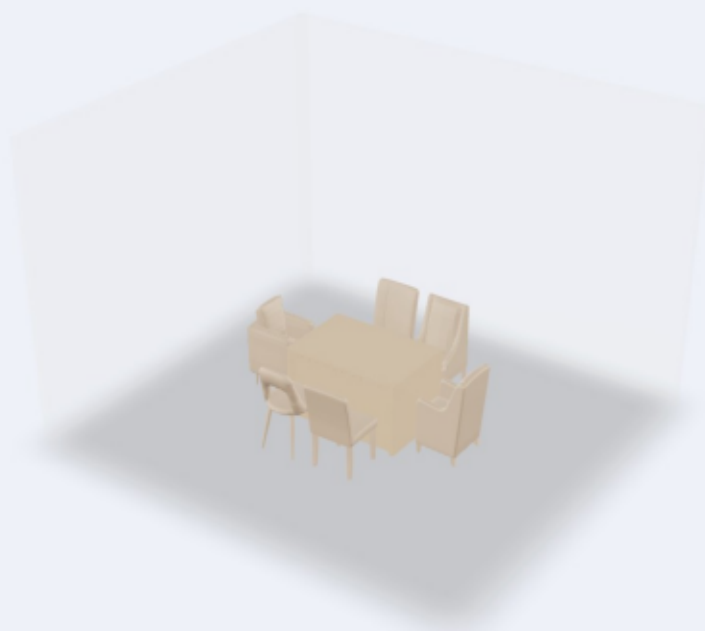
2D plan — clearance zones + ASR panel

Dining — Esszimmer — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Dining — Esszimmer — 7 pieces (X-ray)



# Kitchen — Küche

5 × 4 m · medium density · 13 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 4 × **chair**

Regular chair — at tables (max 4 per side, centered); spares park at the wall facing the room

### 3 × **cabinet**

Perimeter storage — flat against a wall, doors openable

### 1 × **fridge**

Kitchen/break perimeter appliance

### 1 × **waste bin**

At arm's reach BESIDE the desk/table (~0.2 m) — movable, minimal clearance

### 1 × **table**

Shared surface — chairs gather around it; rectangular tables seat the edges, never a 'petal' ring

### 1 × **microwave**

ON a counter or table — never on the floor

### 1 × **coffee machine**

ON a counter or table

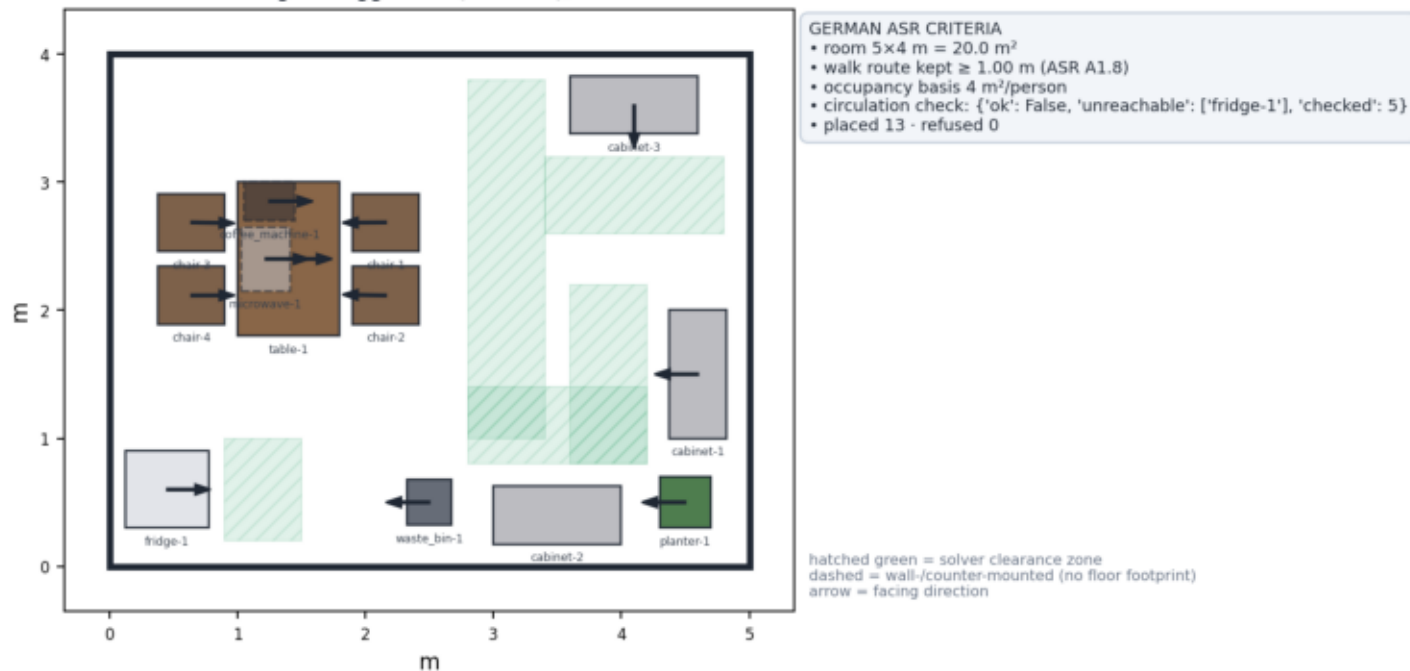
### 1 × **planter**

Corners and window zones only — never mid-room

# Kitchen — Küche

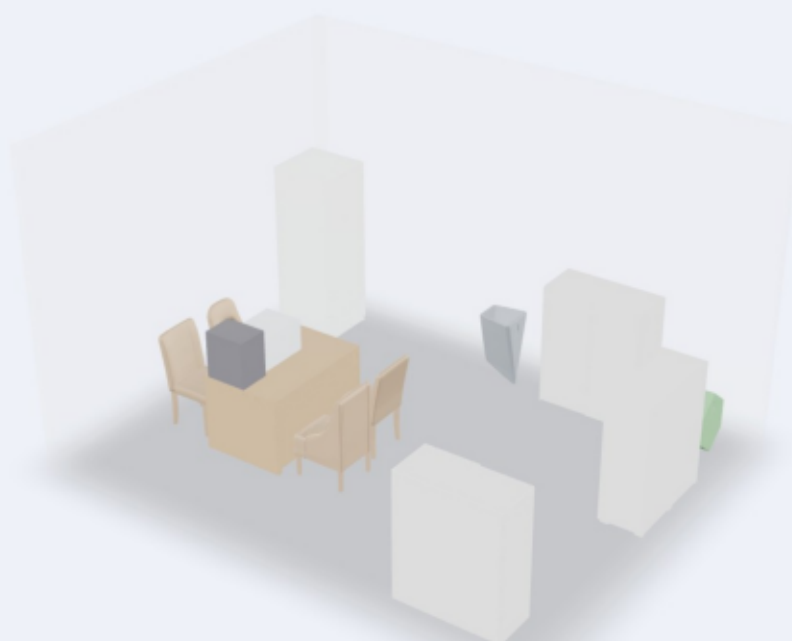
## 2D plan — clearance zones + ASR panel

Kitchen — Küche — engine suggestion (medium), clearance zones + ASR



## 3D X-ray — shell ghosted, real placed pieces

Kitchen — Küche — 13 pieces (X-ray)



# Storage — Lager/Keller

4 × 3 m · medium density · 2 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 1 × **bookshelf**

Perimeter storage — flat against a wall

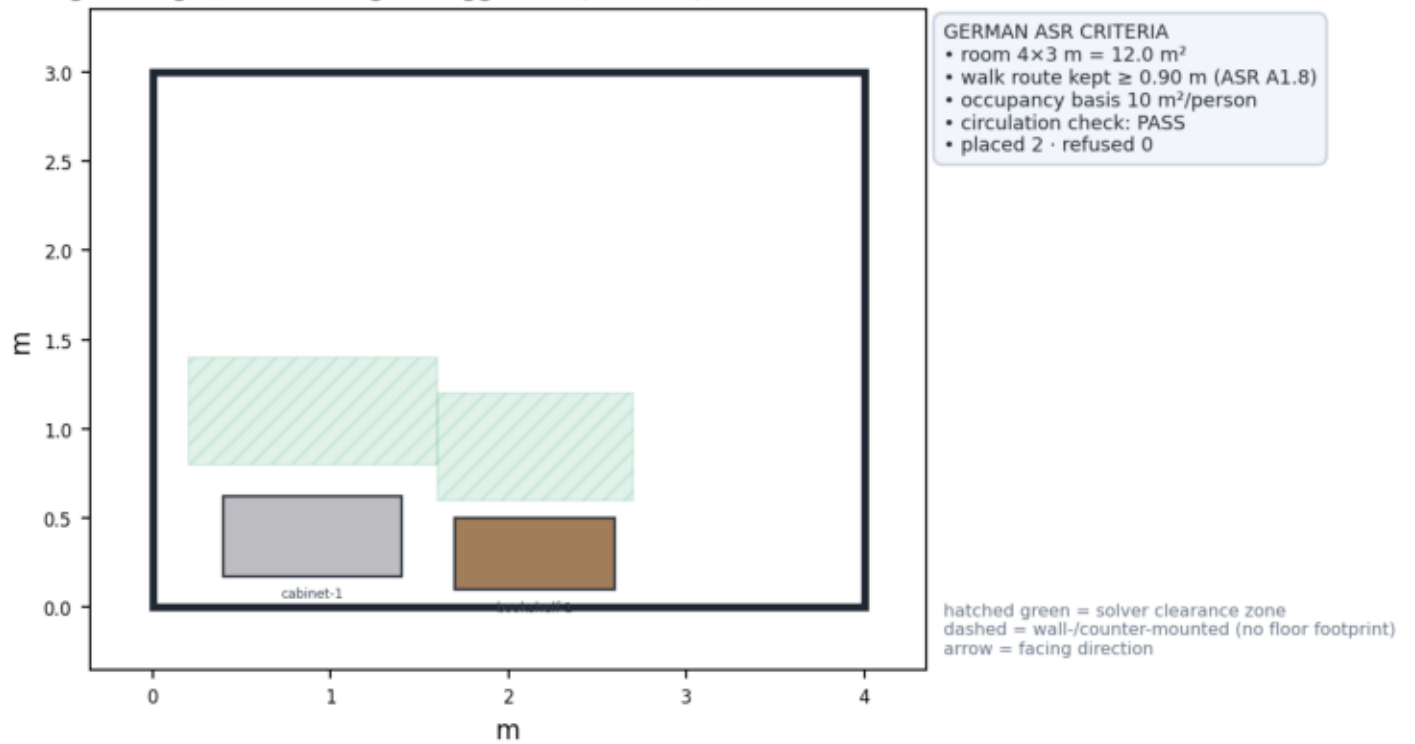
### 1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

# Storage — Lager/Keller

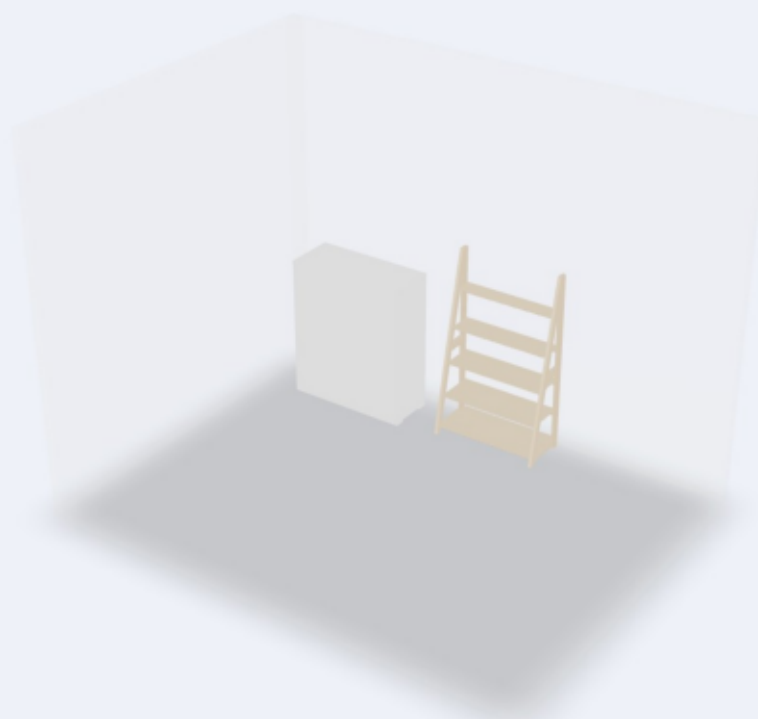
2D plan — clearance zones + ASR panel

Storage — Lager/Keller — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Storage — Lager/Keller — 2 pieces (X-ray)



# Wardrobe — Garderobe

4 × 3 m · medium density · 6 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **coat rack**

Near the door / entrance zone

### 2 × **locker**

Wall line-up, typically entrance side

### 1 × **mirror**

Wall-mounted, entrance/wardrobe zone

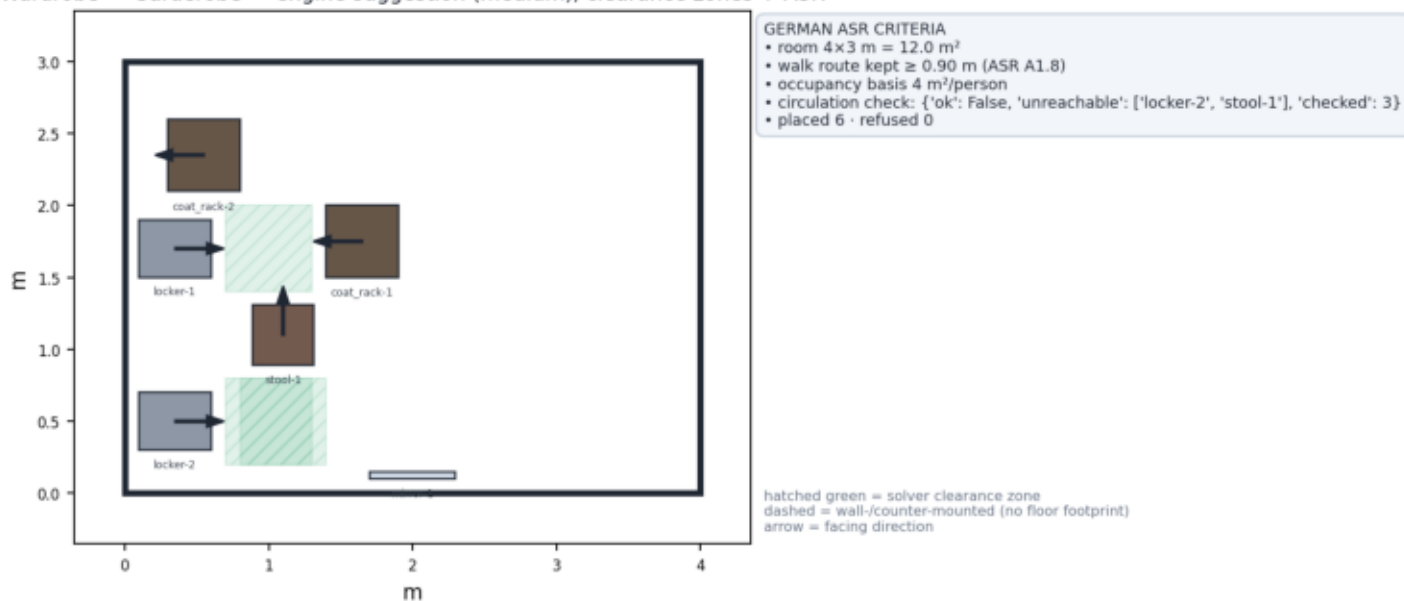
### 1 × **stool**

Light seating at small tables / bar height

# Wardrobe — Garderobe

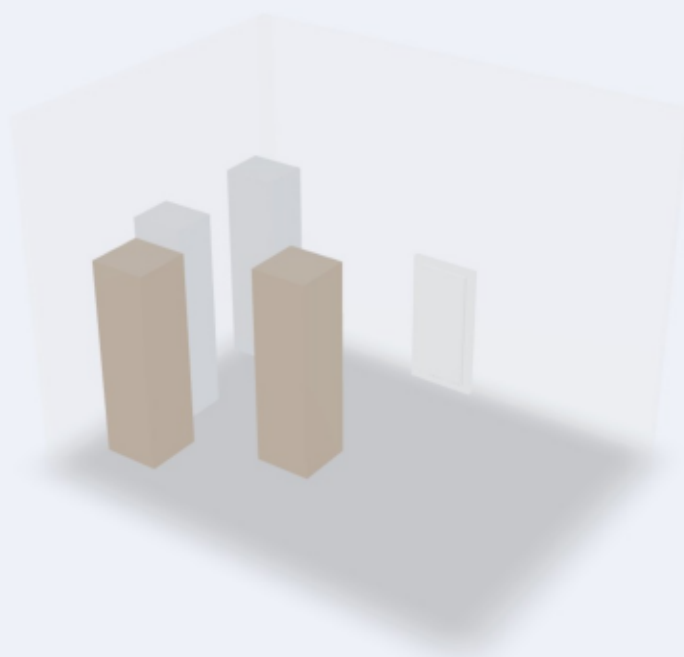
## 2D plan — clearance zones + ASR panel

Wardrobe — Garderobe — engine suggestion (medium), clearance zones + ASR



## 3D X-ray — shell ghosted, real placed pieces

Wardrobe — Garderobe — 6 pieces (X-ray)



# Entry — Foyer/Eingang

5 × 3 m · medium density · 7 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **planter**

Corners and window zones only — never mid-room

### 2 × **armchair**

In ROWS: 2 face the same way; 4 = two opposed pairs; never blocking the walkway

### 1 × **coat rack**

Near the door / entrance zone

### 1 × **mirror**

Wall-mounted, entrance/wardrobe zone

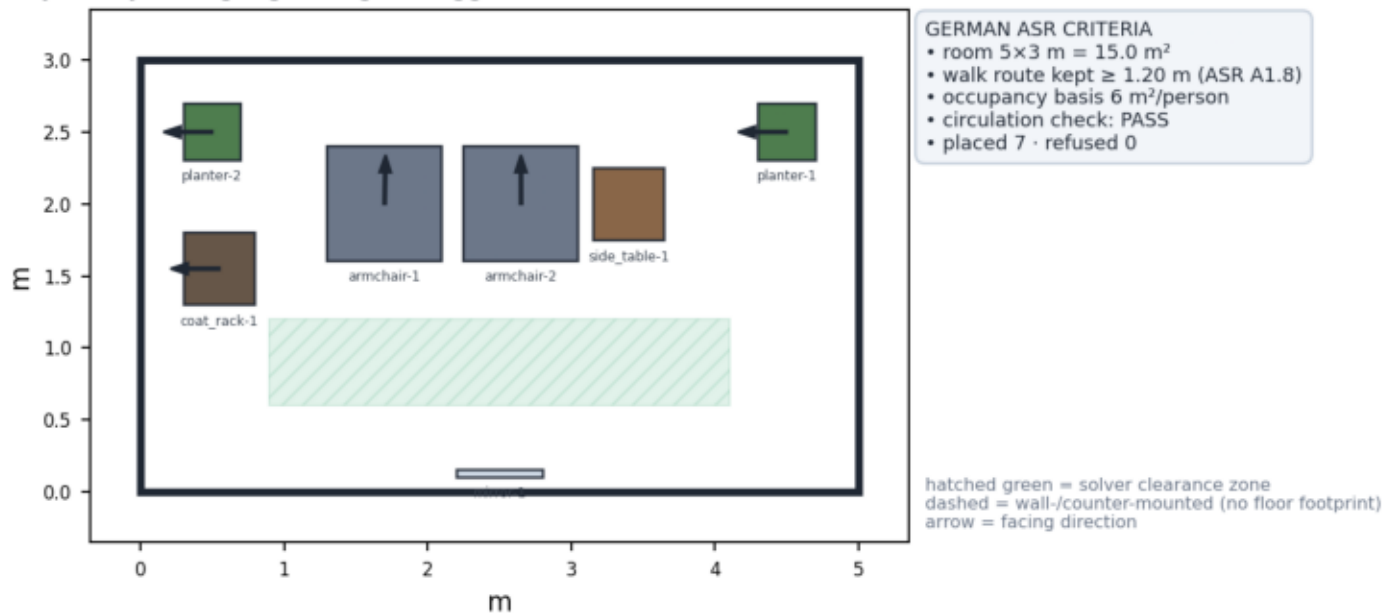
### 1 × **side table**

At the end of a seating row / beside armchairs

# Entry — Foyer/Eingang

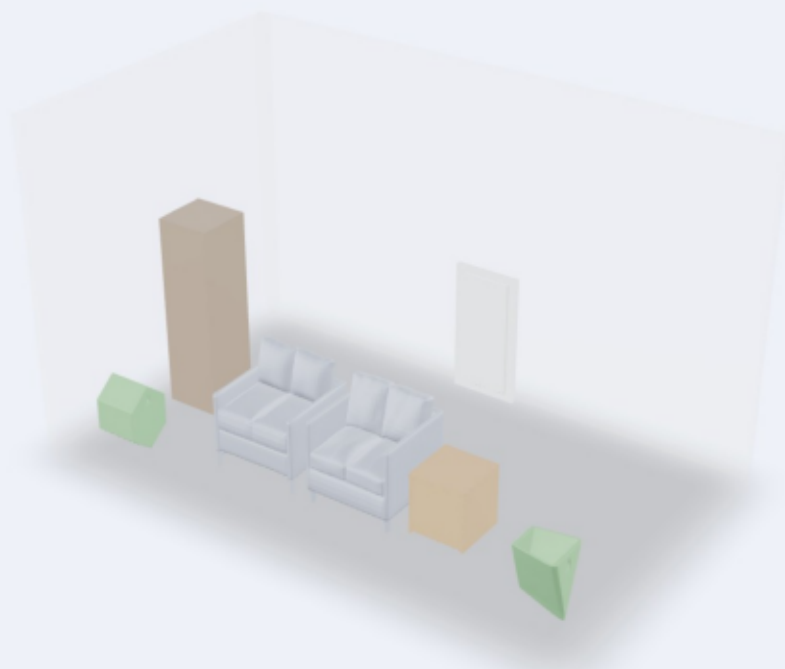
2D plan — clearance zones + ASR panel

Entry — Foyer/Eingang — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Entry — Foyer/Eingang — 7 pieces (X-ray)



# Balcony — Balkon

4 × 2.5 m · medium density · 5 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **planter**

Corners and window zones only — never mid-room

### 2 × **stool**

Light seating at small tables / bar height

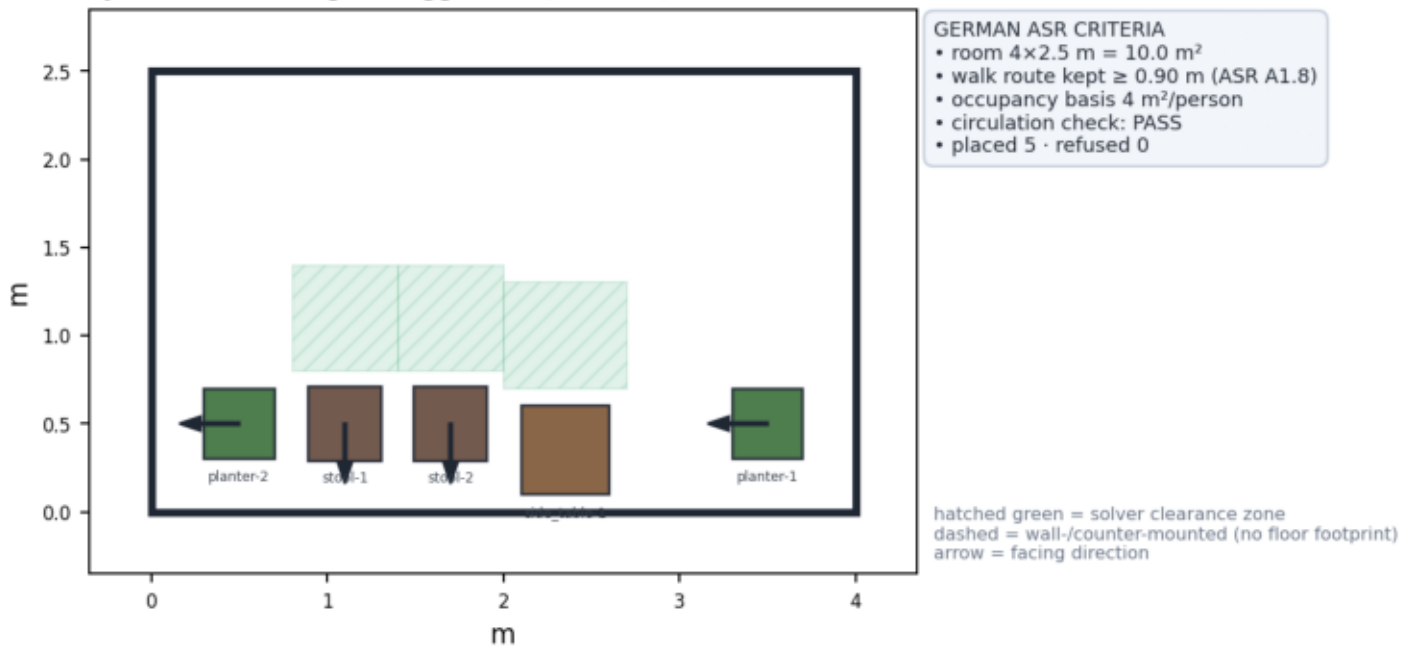
### 1 × **side table**

At the end of a seating row / beside armchairs

# Balcony — Balkon

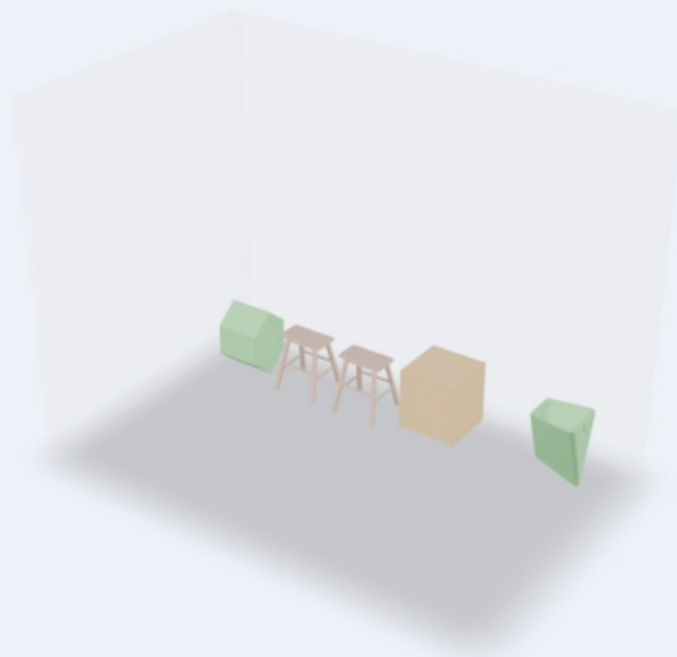
2D plan — clearance zones + ASR panel

Balcony — Balkon — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Balcony — Balkon — 5 pieces (X-ray)



# Print room — Kopierraum

3.5 × 3 m · medium density · 5 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **printer**

Shared wall position, reachable from the aisle

### 1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

### 1 × **waste bin**

At arm's reach BESIDE the desk/table (~0.2 m) — movable, minimal clearance

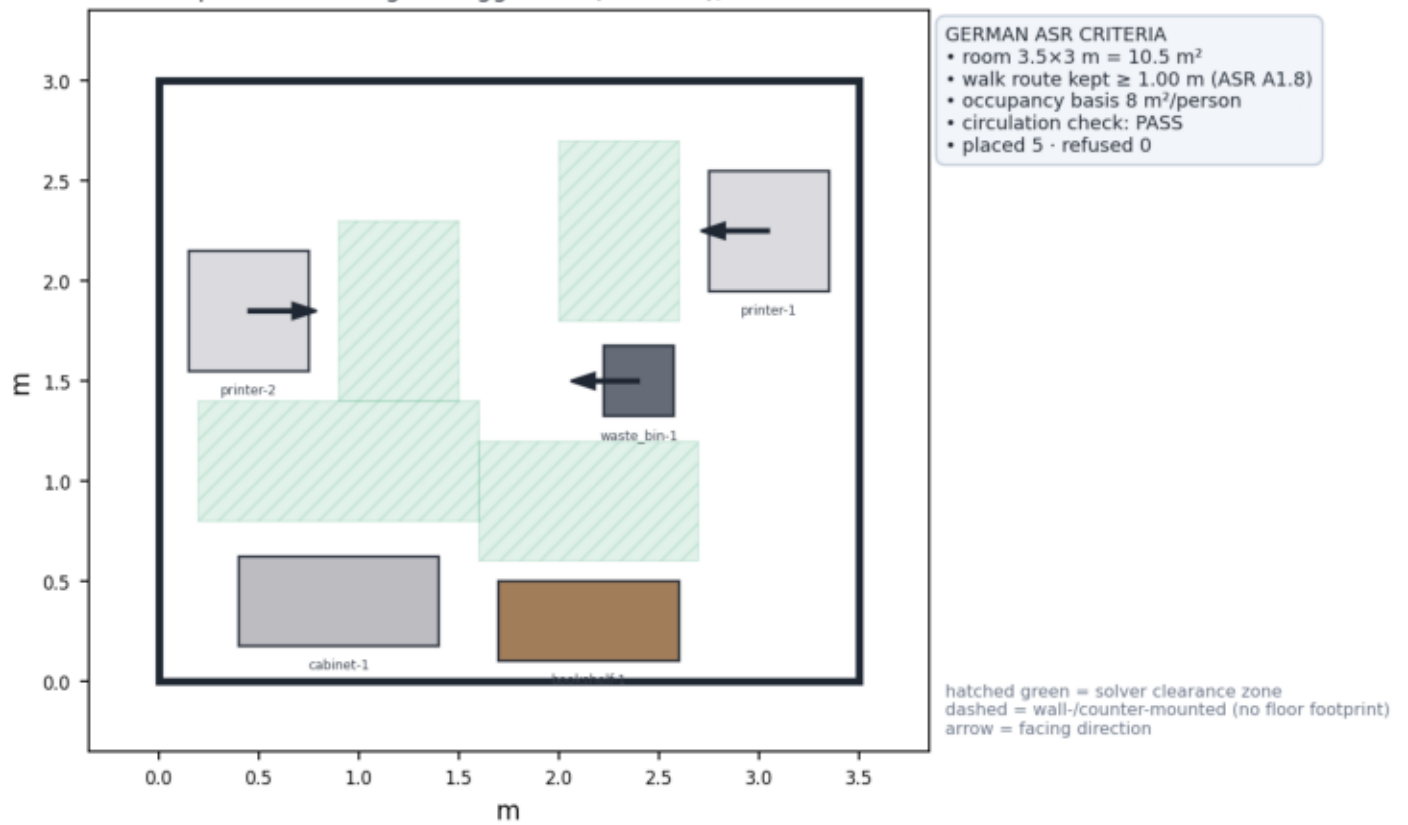
### 1 × **bookshelf**

Perimeter storage — flat against a wall

# Print room — Kopierraum

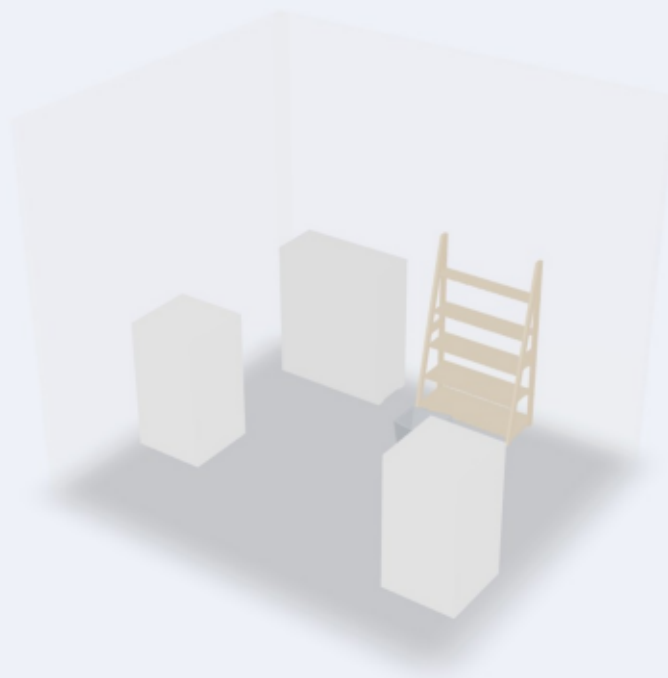
2D plan — clearance zones + ASR panel

Print room — Kopierraum — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

Print room — Kopierraum — 5 pieces (X-ray)



# IT/server — EDV-Raum

4 × 3.5 m · medium density · 4 items placed · 0 refused

## WHAT THE ENGINE PUT IN THIS ROOM — AND WHY

### 2 × **server rack**

IT room walls, service aisle  $\geq 1.0$  m kept free

### 1 × **cabinet**

Perimeter storage — flat against a wall, doors openable

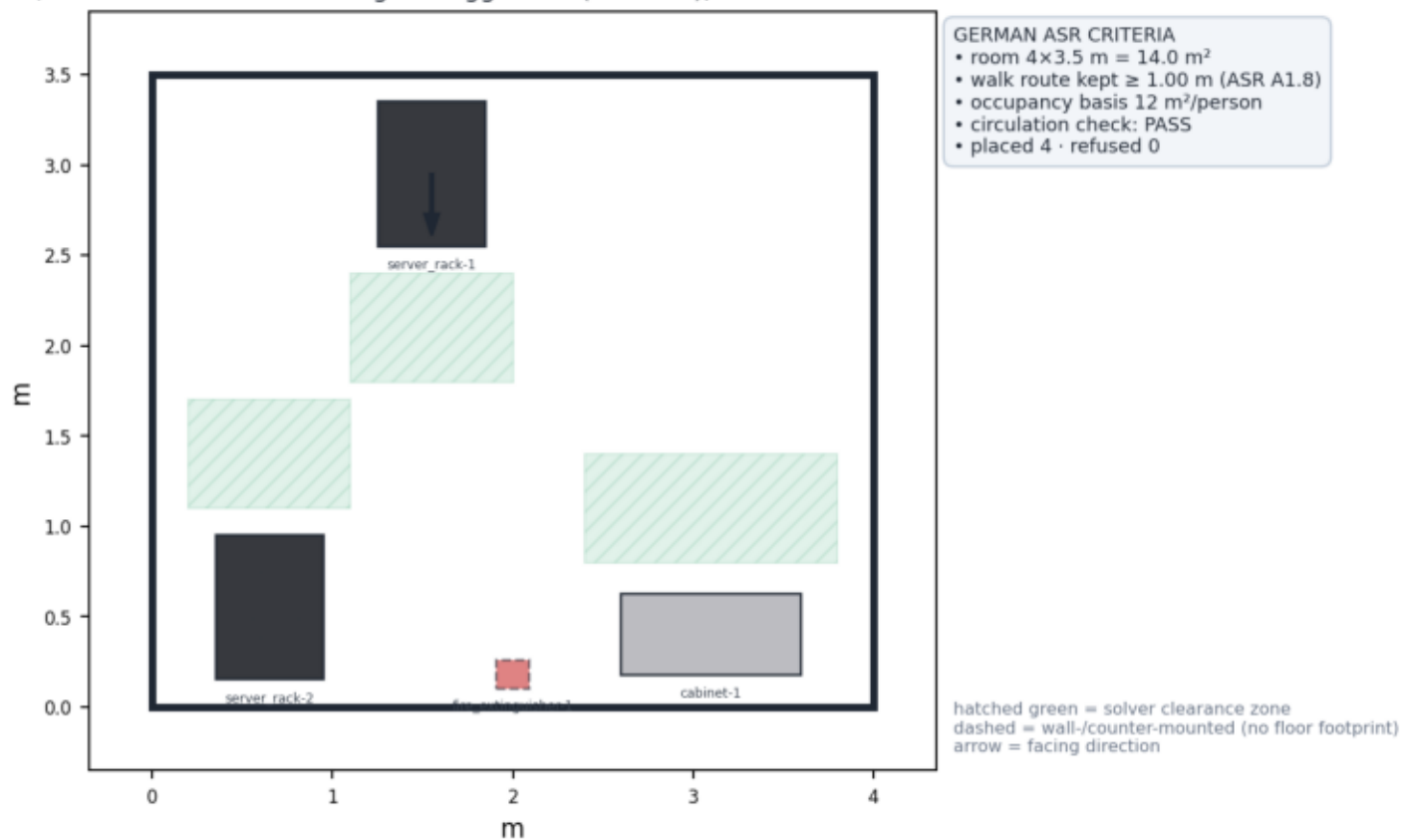
### 1 × **fire extinguisher**

Wall-mounted 1.00 m, PROTECTED access strip (ASR A2.2)

# IT/server — EDV-Raum

2D plan — clearance zones + ASR panel

IT/server — EDV-Raum — engine suggestion (medium), clearance zones + ASR



3D X-ray — shell ghosted, real placed pieces

IT/server — EDV-Raum — 4 pieces (X-ray)

